

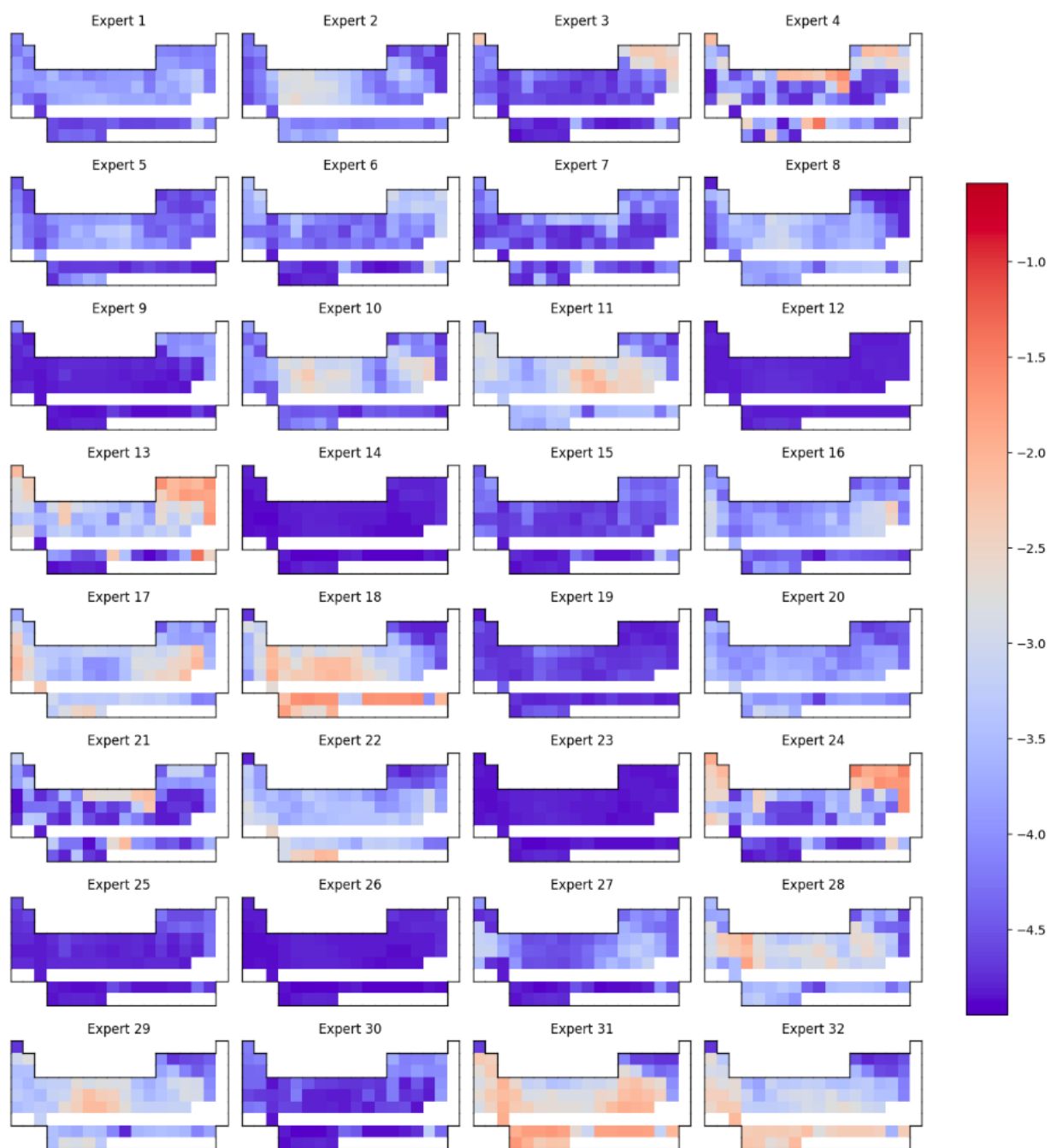


Figure 5 Log mean expert coefficient across element-expert pairs.

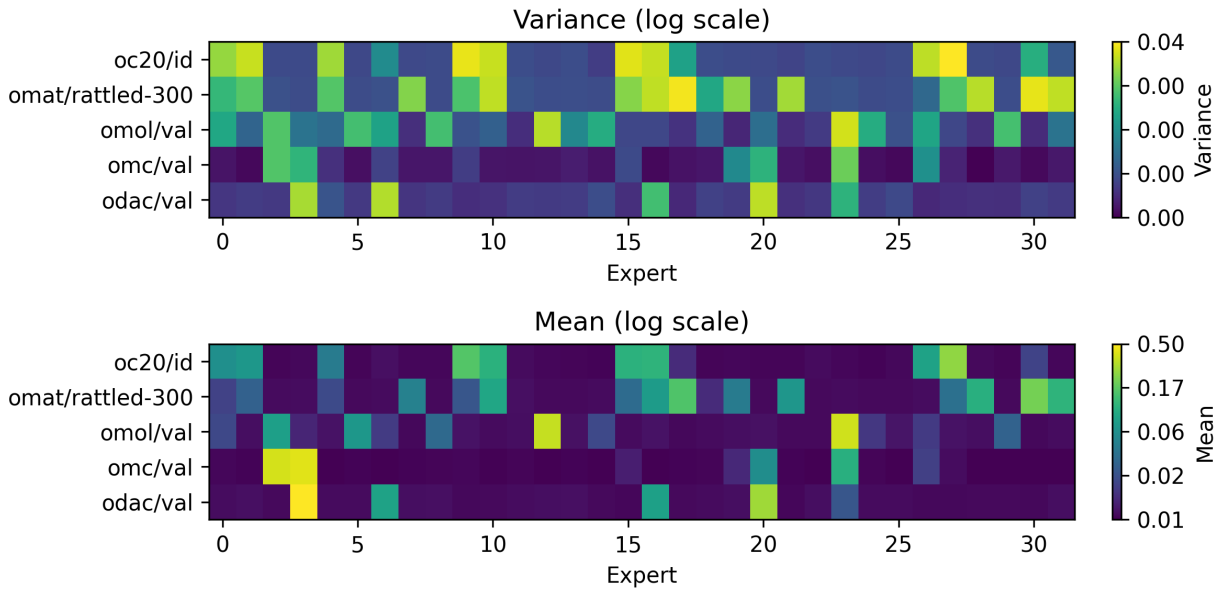


Figure 6 Log mean expert coefficient across element-expert pairs.

G Diatomics

The potential energy curves of diatomic molecules serve as standard unit tests in computational chemistry, albeit a challenging one even for quantum chemistry methods like DFT. Previous versions of UMA (1 and 1.1) struggled with this task and showed unphysical behavior at times. In particular, the OMat task produced large energy fluctuations beyond the bonding region (Figure 7). It was unclear if these issues originated from the model, the data, or the training procedure.

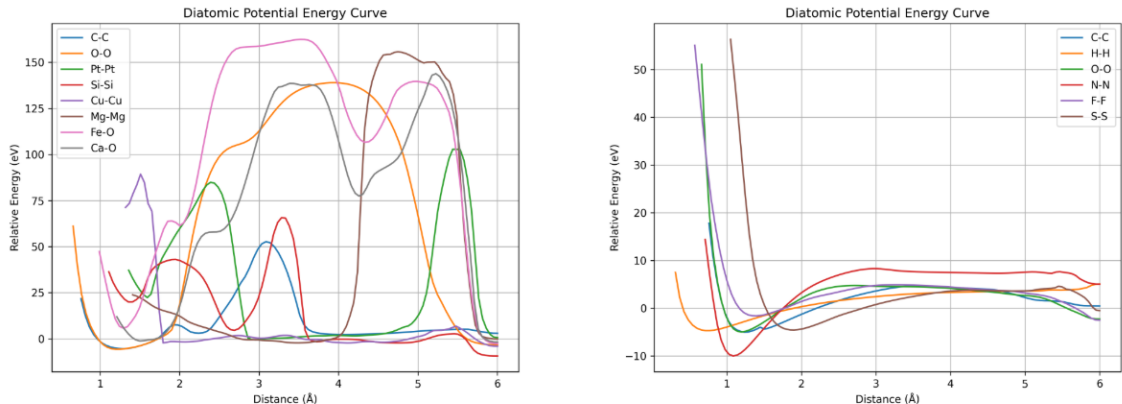


Figure 7 Diatomic curves for UMA-S-1.1. Left is the OMat task, right is the OMol task.

Smooth diatomics

With the updated UMA-1.2 training procedure and data ratios, we demonstrate in Figure 8 that the OMat and OMol diatomics are qualitatively smooth. This unreleased version of UMA-1.2 was trained without any diatomic data.

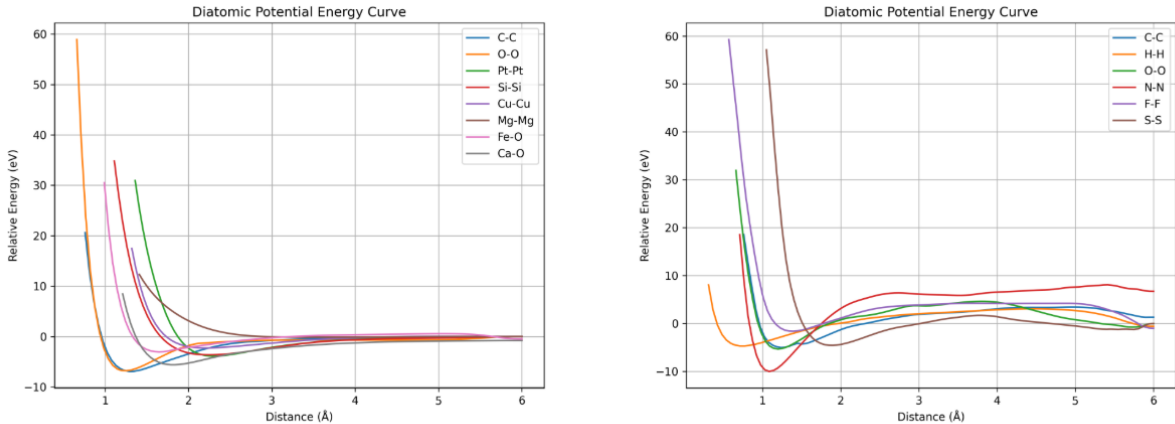


Figure 8 Diatomic curves for UMA-S-1.2 (unreleased version trained without diatomic data added). Left is the OMat task, right is the OMol task.

Smooth and accurate diatomics

While the diatomic curves in Figure 8 are smooth, the accuracy of the predictions are lacking (Table 26). To have both smooth and accurate predictions, and to produce the best model possible we trained another version of UMA-1.2 which included diatomic data for OMat and OMol. The two models were trained identically with the exception of the additional diatomic data. Not surprisingly as shown in Table 26, the addition of diatomic

data significantly improves the accuracy of diatomic energy and forces estimates. The model trained with diatomic data is released publicly.

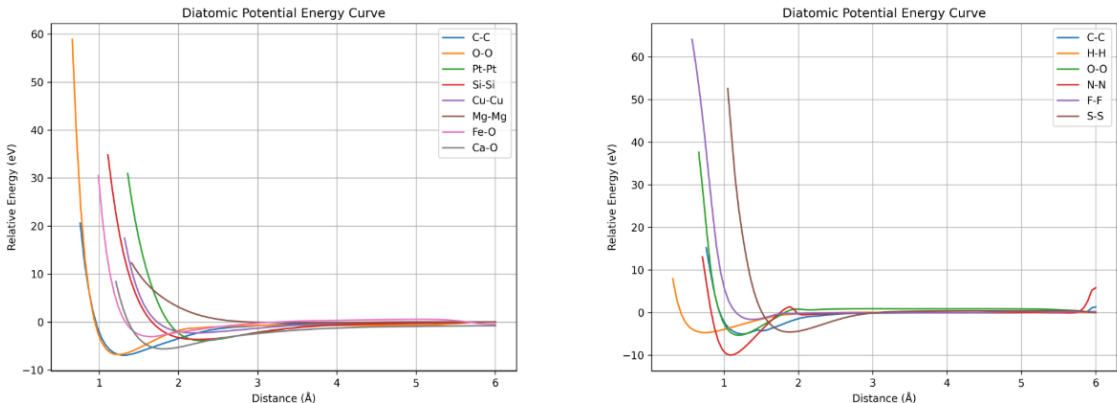


Figure 9 Diatomic curves for UMA-S-1.2 (released version trained with OMat and OMol diatomic data added), left is OMat task, right is OMol task.

Table 26 Diatomic energy (meV) and force (meV/Å) MAE for OMat and OMol tasks.

Model	OMat E MAE	OMat F MAE	OMol E MAE	OMol F MAE
UMA-S-1.2 without diatomic data	128.6	279.6	2429.5	1326.9
UMA-S-1.2 with diatomic data	37.8	52.5	60.5	103.4

Limitations

For the OMol task, some issues with the asymptotic behavior of the diatomic curves remain but this is a fundamental limitation of the way global charge/spin are embedded and how this interacts with the envelope function (which smoothly decay edges to zero at the cutoff). Once the pair of atoms reach a separation of $\sim 5.5\text{\AA}$ the envelope function decays the edge interaction towards zero resulting in essentially two disconnected atoms and given a global charge and spin the model thinks both atoms have the same global charge and spin, which is not generally valid. That is, for N₂ dissociation to take and example, the model beyond 5.5Å is trying to simulate two separate N atoms, each with a spin of 1 (the overall global spin for N₂), but a singlet N atom is unphysical. Additionally, there are a few erroneous features like an small barrier in N-N dissociation around $\sim 1.8\text{-}1.9\text{\AA}$, this is likely caused by data that exists in the training set.